

VIEWS IN SQL: View is database object which is like table but logical. We can call it as a logical or virtual table because it does not has a physical existence.

- It is a logical table use to get the required information from the table. View will be created by using select statement and table used for the creation of the view is called as base table.
- View will not store records in it and will not occupy memory space with help of structure existing in it and records will be displayed from table.

- View is logical representation or virtual representation .it is a dependent where as table an independent is because view is extracted from the table.
- If we want to access the data from the table it's not necessary to change the data direct to the table but we can access by having a view.
- Views are used for security purpose in databases, views restricts the user from viewing certain column and rows means by using view we can apply the restriction on accessing the particular rows and columns for specific user.
- Views display only those data which are mentioned in the query, so it shows only data which is returned by the query that is defined at the time of creation of the View.

Why We Need Views: To protect the data. If you have a table containing sensitive data in certain columns, you might wish to hide those columns from certain groups of users. For instance, customer names, addresses and their social security numbers might all be stored in the same table; however, for lower level employees like shipping clerks, you can create a view that only displays customer name and address. You can grant permissions to a view without allowing users to query the original tables.

- A view is a logical table but what it stores internally is a select statement that is used for creating the view. So that whenever a user performs any operation on the view like select, insert, update or delete internally the view performs those operations on a table.
- Simply we can say that view will act as an interface between the data provider (Table) and the User.

View is created based on a table any changes that are performed on the table reflects into the view any changes performed on the view reflect into the table also.

View is classified into two types. These are

- Simple view(Updatable view)
- Complex view(Non-Updatable view)

Simple view: we create a view based on one table is called simple view or Updatable view.

Complex view: we create a view based on more than one table is called complex view or Non-Updatable view.

Syntax: create view <view name> as select *

from <table name> **Ex1:** create view **simpleview** as

select * from **emp**;

Ex2: CREATE VIEW **COMPLEXVIEW** AS SELECT
E.EID,E.ENAME,D.DNO,D.DNAME FROM **EMP** E INNER JOIN DEPT D
ON E.DEPTNO=D.DEPTNO